IDAT7215
Computer Programming for Product Development and Applications

Lecture 2-2: Python Libraires: Matplotlib

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Outline

- Why Data Visualization
- What is Data Visualization
- Introduction to Matplotlib
- Types of Plots
  - Line Plot
  - Bar Graph
  - Histograms
  - Scatter Plot
  - Area Plot
  - Pie Plot

Why Data Visualization?

- Human brain can process information easily when it is in pictorial or graphical form



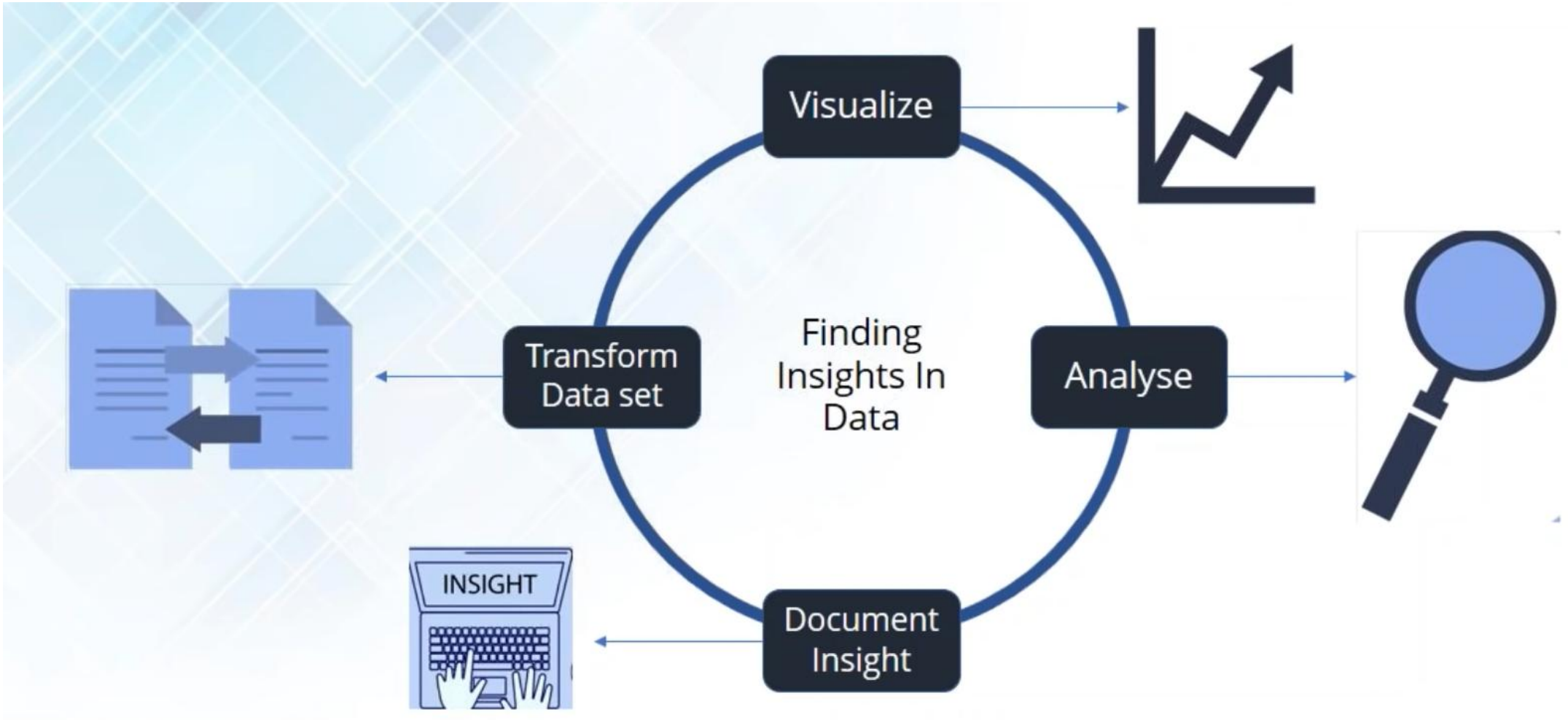
Why Data Visualization?

- Data visualization allows us to quickly interpret the data and adjust different variables to see their effect.



What is Data Visualization?

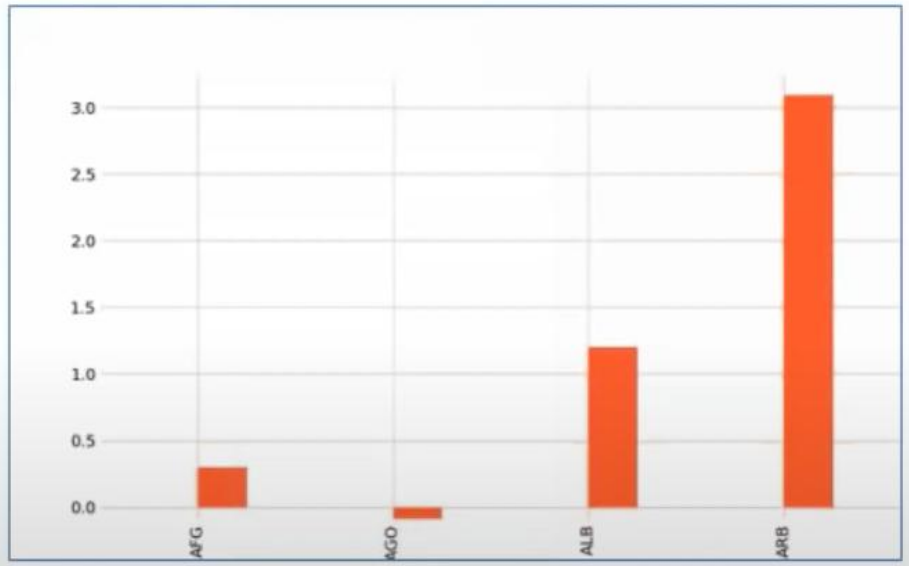
- Data visualization is the presentation of data in a pictorial or graphical format.



What is Matplotlib?

- Matplotlib is a Python package used for 2D graphics.

Country Name	Country Code	2010	2011	2012	2013	2014
Afghanistan	AFG	20.6	20.9	19.7	21.1	20.8
Angola	AGO	10.8	10.7	10.7	10.6	10.5
Albania	ALB	25.799999	27	28.3	28.7	29.2
Arab World	ARB	25.022214	28.11752	29.11321	29.33531	29.70457
United Arab Emirates	ARE	9.8000002	9.8	9.8	9.9	10
Argentina	ARG	19.5	18.8	18.4	19.7	21.3
Armenia	ARM	38.299999	38.7	35	32.5	35.1
Australia	AUS	11.4	11.4	11.7	12.2	13.1
Austria	AUT	8.8000002	8.2	8.7	9.1	9.2
Azerbaijan	AZE	14.6	14.5	14.3	13.4	13.6
Burundi	BDI	10.8	10.8	10.8	10.8	10.7
Belgium	BEL	22.5	18.6	19.7	23.1	23.6
Benin	BEN	2	2	2	1.8	1.7
Burkina Faso	BFA	5.1999998	5.3	5.2	5.2	5
Bangladesh	BGD	8.1999998	8.2	8.2	8.9	9.1
Bulgaria	BGR	22.9	25.2	28.2	29.7	25.9
Bahrain	BHR	10.2	11.4	10.5	10.6	10.9
Bahamas, The	BHS	36	27.2	30.4	30.8	30.1
Bosnia and Herzegovina	BIH	57.200001	57.1	61.7	57.4	57.5
Belarus	BLR	13.2	12.5	11.8	12	12
Belize	BLZ	20.9	24.3	26	22.4	22



What is Matplotlib?

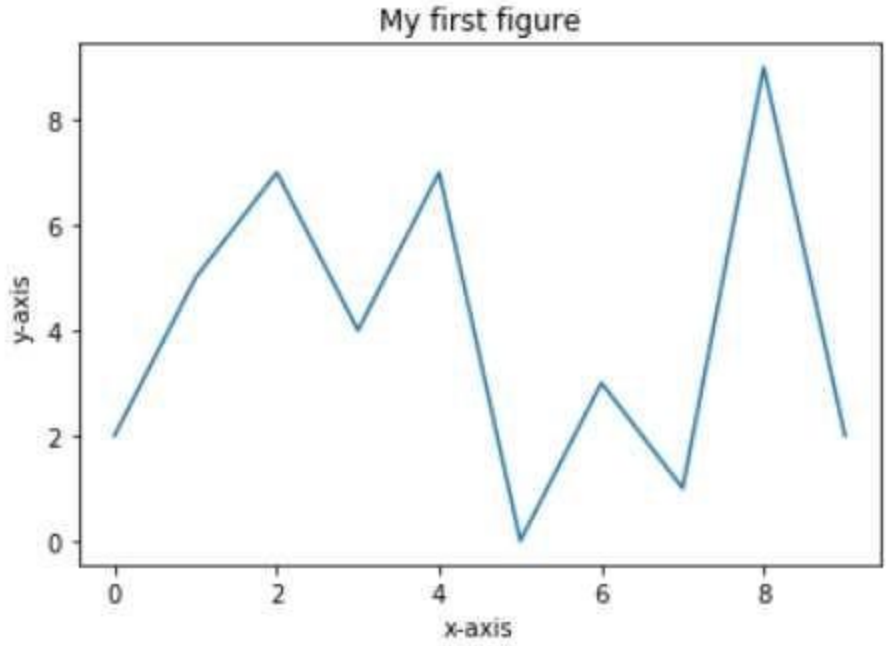
- Matplotlib is an amazing visualization library in python for 2D plots of arrays.
- Matplotlib is a multi-platform data visualization library built on NumPy arrays and designed to work with broader SciPy stack. It was **introduced by John Hunter in the year 2002**.
- One of the greatest benefits of visualization is that it allows us visual access to huge amount of data in easily digestible visuals.
- Matplotlib is the most common low-level visualization library for Python.
- It can create line graphs, bar plots, scatter plots, density plots, histograms, heatmaps, and so on.

Simple Figure

- Simply line plot

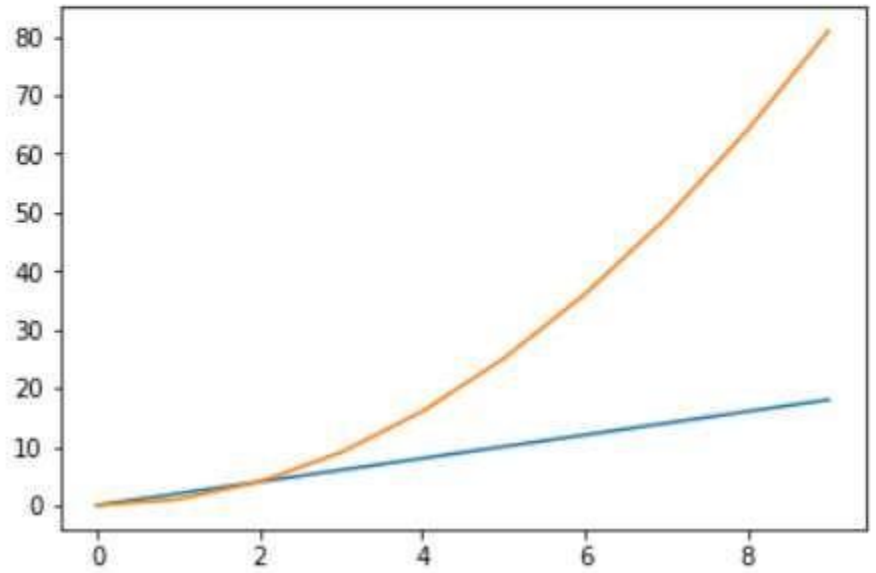
```
import numpy as np
import matplotlib.pyplot as plt
# add IPython magic command %matplotlib inline
# to print pyplot output inline in the notebook withput calling plt.show()
%matplotlib inline

a=np.array([2, 5, 7, 4, 7, 0, 3, 1, 9, 2])
plt.plot(a)                    # plot the points in the array a
plt.title("My first figure")  # Add a title to the figure
plt.xlabel("x-axis")         # Give a label to the x-axis
plt.ylabel("y-axis")         # Give a label to the y-axis
plt.show()                      # Tell matplotlib to output the figure.
                                # Not strictly required in notebooks (but a bit neater).
```



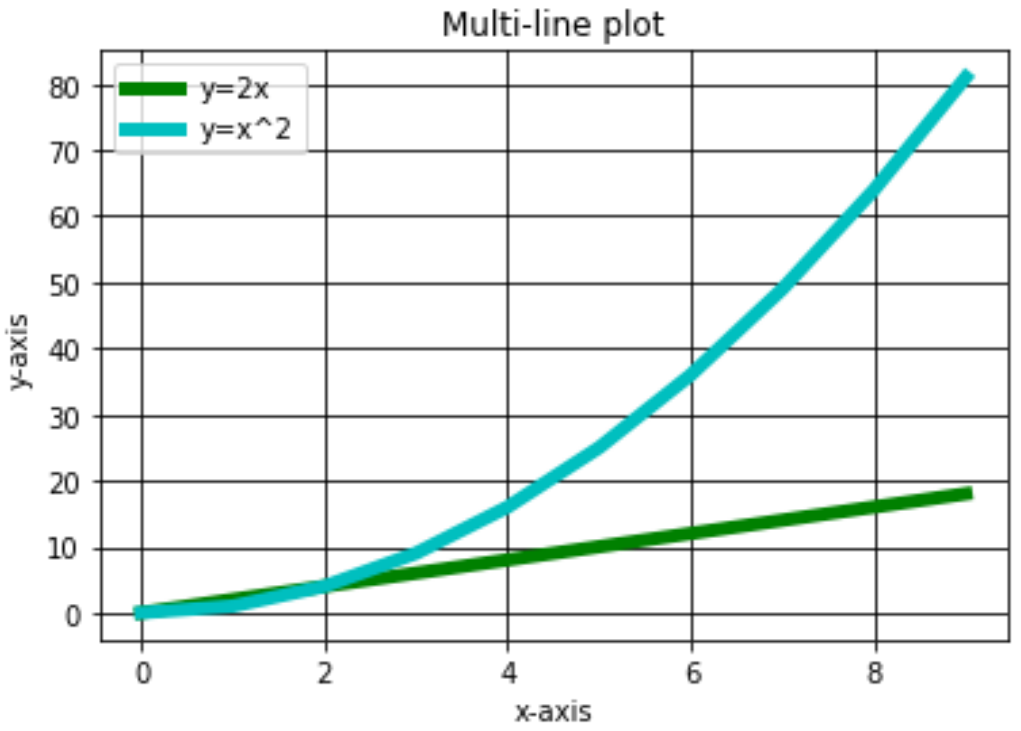
Plotting two lines in the same figure

```
# if plt.plot(*args) are run in the same cell
# we will plot on the same axes
x = np.arange(10)
plt.plot(x,2*x)
plt.plot(x,x**2)
plt.show()
```



Plotting two lines in the same figure

```
# If plt.plot(*args) are run in the same cell
# we will plot on the same axes
x = np.arange(10)
y1=2*x
y2=x**2
plt.plot(x,y1,'g',label='y=2x',linewidth=5)
plt.plot(x,y2, 'c',label='y=x^2',linewidth=5)
plt.title('Multi-line plot')
plt.xlabel('x-axis')
plt.ylabel('y-axis')
plt.legend()
plt.grid(True, color='k')
plt.show();
```

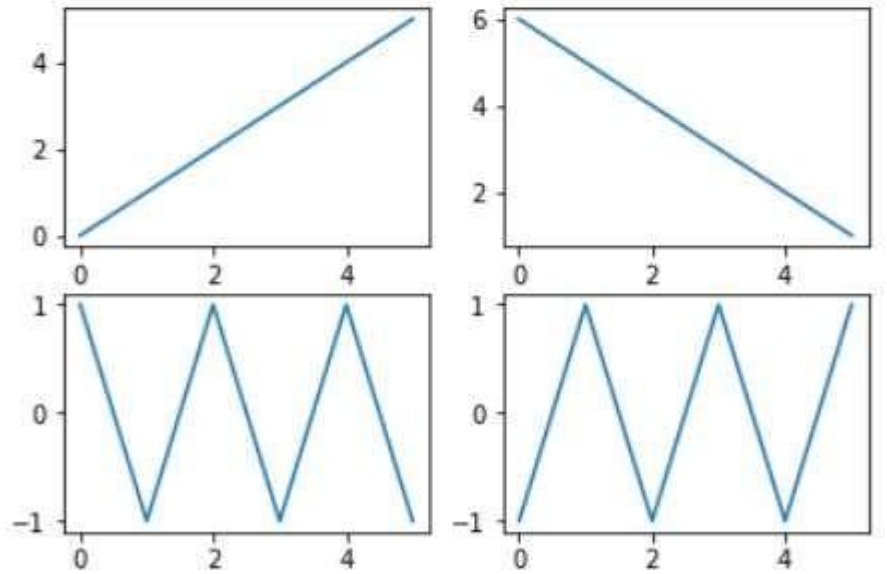


Sub-figures

- One can create a figure with several subfigures using the command `plt.subplots`.
- It creates a grid of subfigures, where the number of rows and columns in the grid are given as parameters.
- It returns a pair of a figure object and an array containing the subfigures. In matplotlib the subfigures are called `axes`.

```
fig, ax = plt.subplots(2,2)
print(ax.shape)
ax[0,0].plot(np.arange(6))           # top left
ax[0,1].plot(np.arange(6,0,-1))      # top right
ax[1,0].plot((-1)**np.arange(6))     # bottom left
ax[1,1].plot((-1)**np.arange(1,7))   # bottom right
plt.show()
```

```
(2, 2)
```



Bar Graph

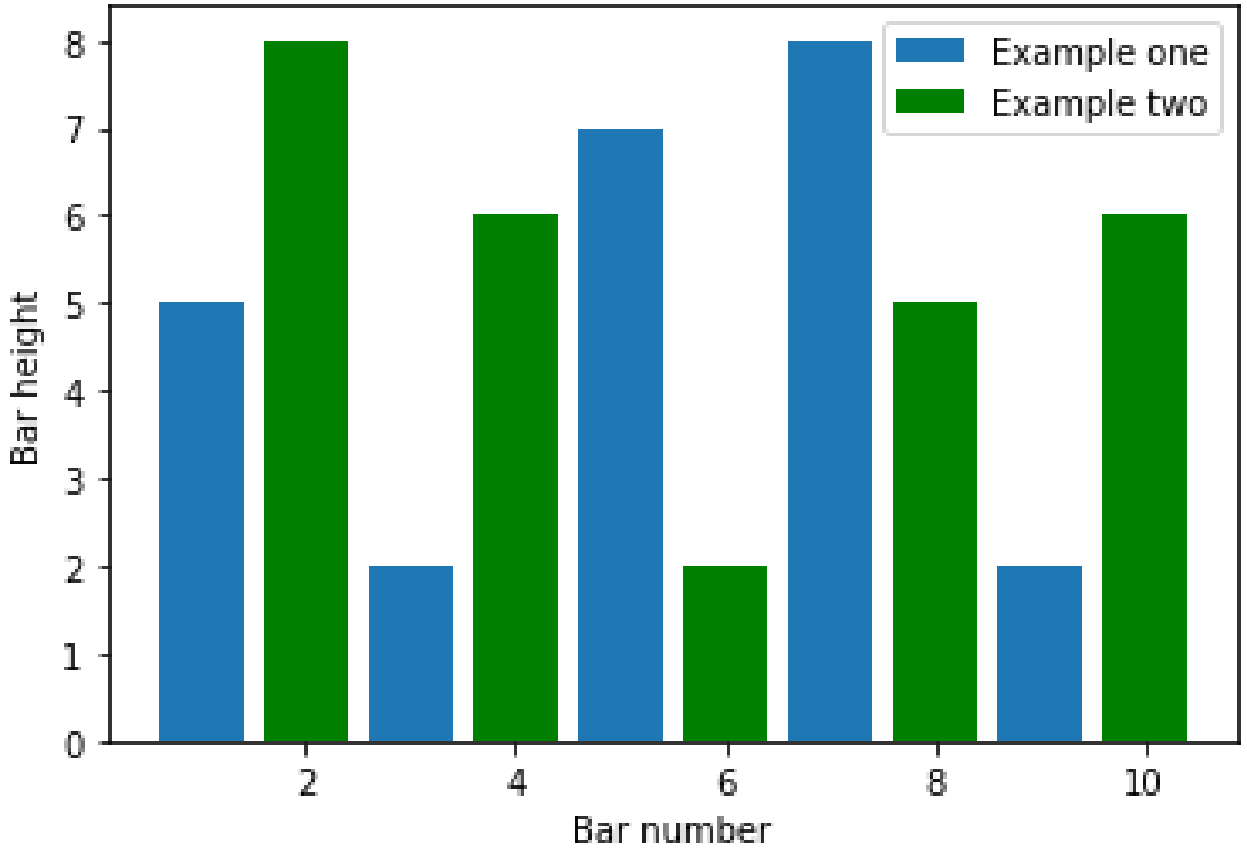
- Bar graph are useful when we want to compare things between groups.
- Useful when changes are larger.

```
import matplotlib.pyplot as plt

x1 = [1,3,5,7,9]   # x-axis values
y1 = [5,2,7,8,2]   # y-axis values

x2 = [2,4,6,8,10]  # x-axis values
y2 = [8,6,2,5,6]   # y-axis values

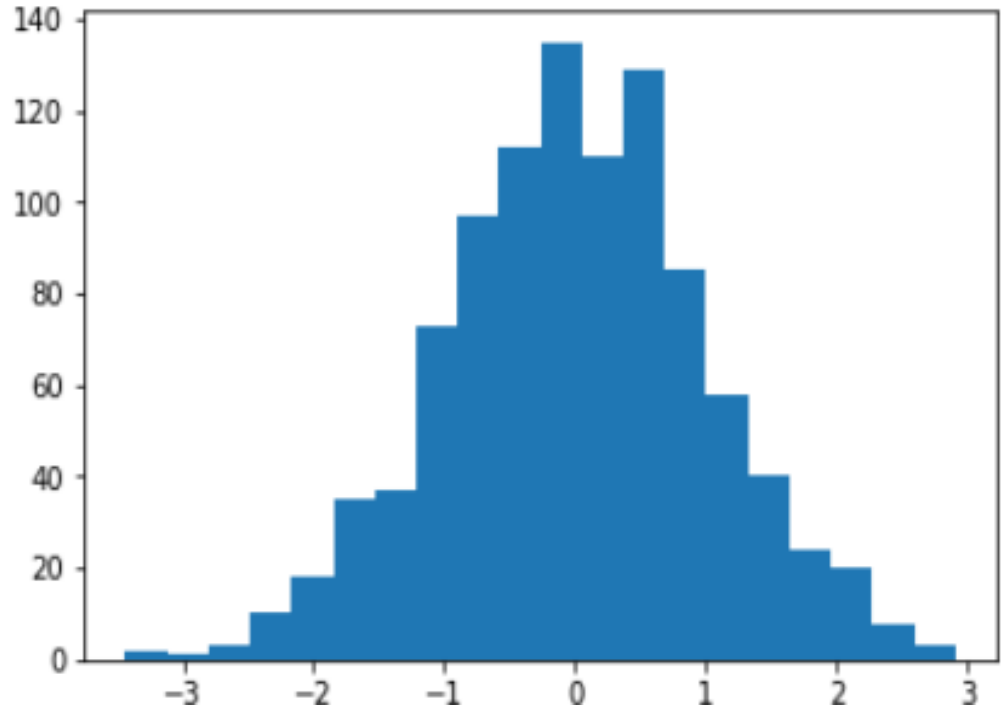
plt.bar(x1,y1, label = "Example one")
plt.bar(x2,y2, label = 'Example two', color='g')
plt.legend()
plt.xlabel('Bar number')
plt.ylabel('Bar height')
plt.show;
```



Histogram

- A ***histogram*** is a way to graphically summarize or describe a data set by visually conveying its distribution using vertical bars.

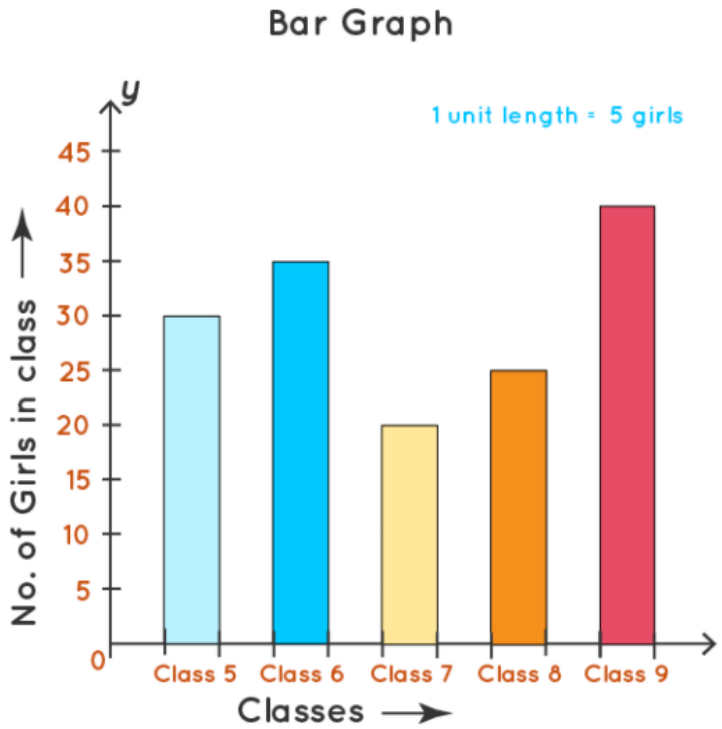
```
x = np.random.randn(1000)
plt.hist(x,20);
```

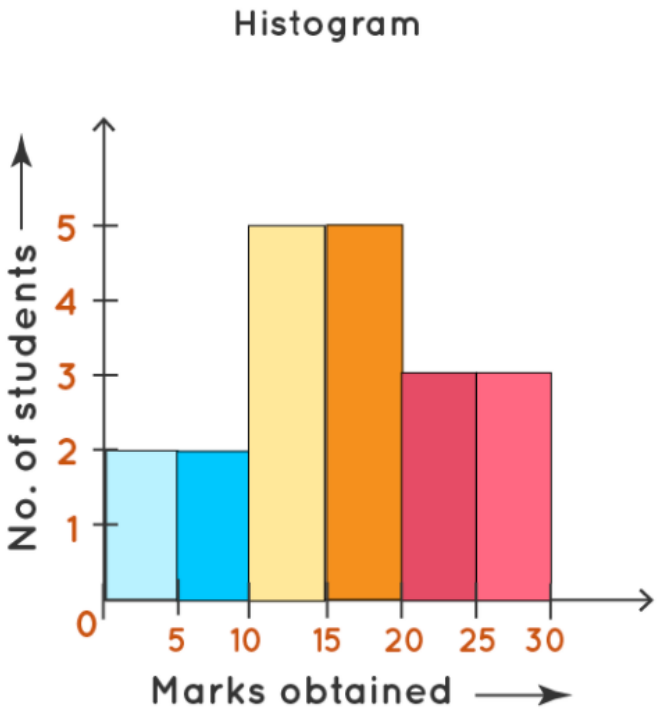


Histogram

- A ***histogram*** is a way to graphically summarize or describe a data set by visually conveying its distribution using vertical bars.

- In histogram we have **quantitative variable** whereas in bar graph we have **categorical variable**.

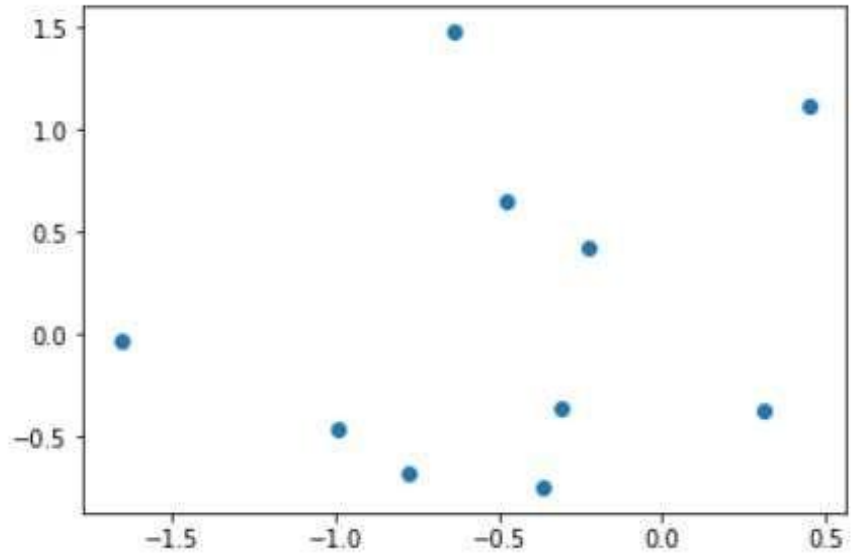




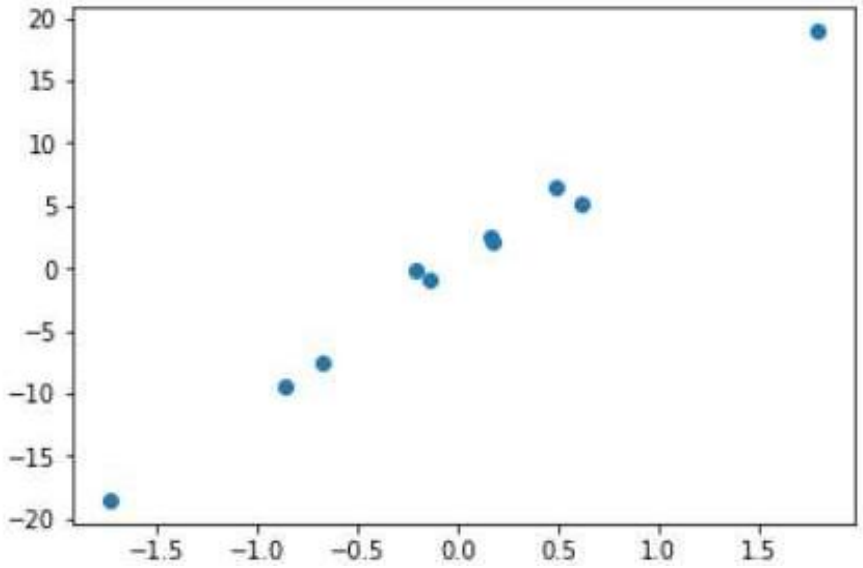
Scatter plots

- The scatterplot is a visualization technique that enjoys widespread use in data analysis and is a powerful way to convey information about the relationship between two variables.

```
x = np.random.randn(10)
y = np.random.randn(10)
plt.scatter(x,y);
```




```
x = np.random.randn(10)
y = x*10 + np.random.randn(10)
plt.scatter(x,y);
```



Area Plot or Stack Plot

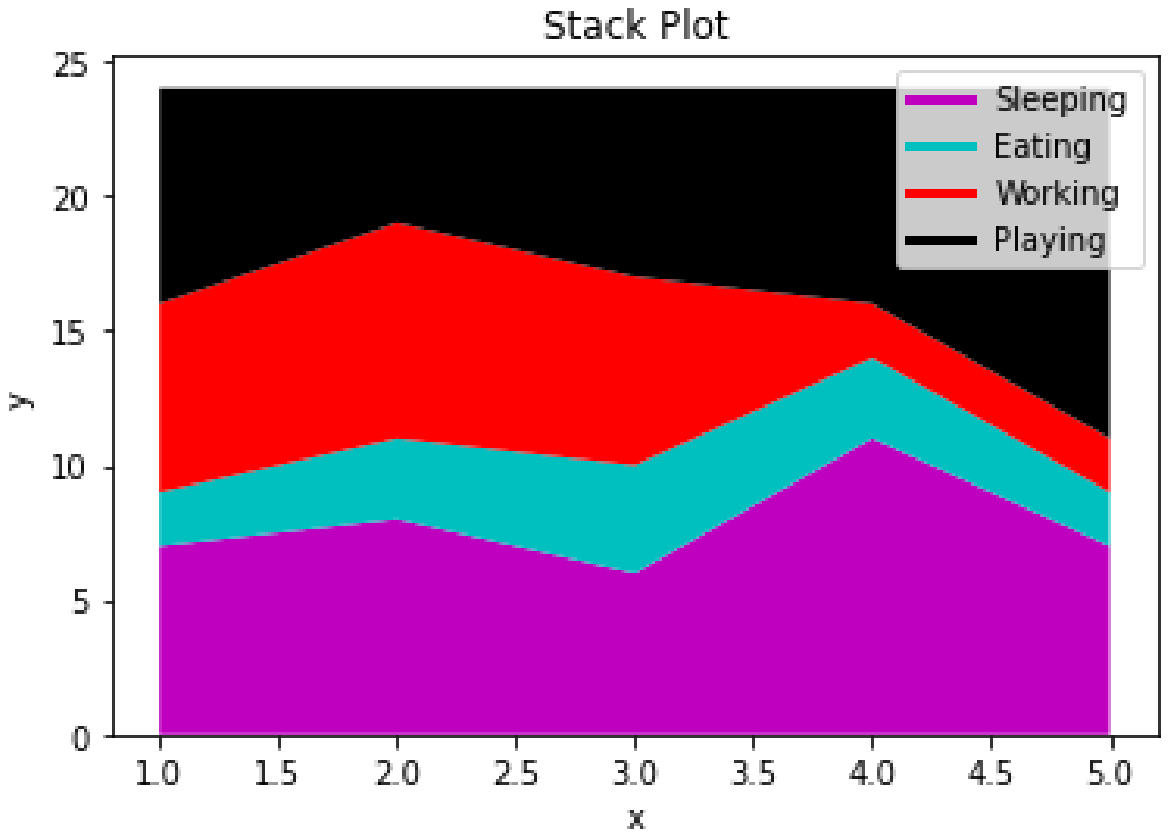
- Area or stack plot are used to draw a stacked area plot. It displays the complete data for visualization.

```
# Data
days = [1,2,3,4,5]
sleeping = [7,8,6,11,7]
eating=[2,3,4,3,2]
working=[7,8,7,2,2]
playing=[8,5,7,8,13]

# Stack plot
plt.stackplot(days,sleeping,eating,working,playing,colors=['m','c','r','k'])

# For labels in legend
plt.plot([],[],color='m',label='Sleeping',linewidth=3)
plt.plot([],[],color='c',label='Eating',linewidth=3)
plt.plot([],[],color='r',label='Working',linewidth=3)
plt.plot([],[],color='k',label='Playing',linewidth=3)

plt.xlabel('x')
plt.ylabel('y')
plt.title('Stack Plot')
plt.legend()
plt.show;
```



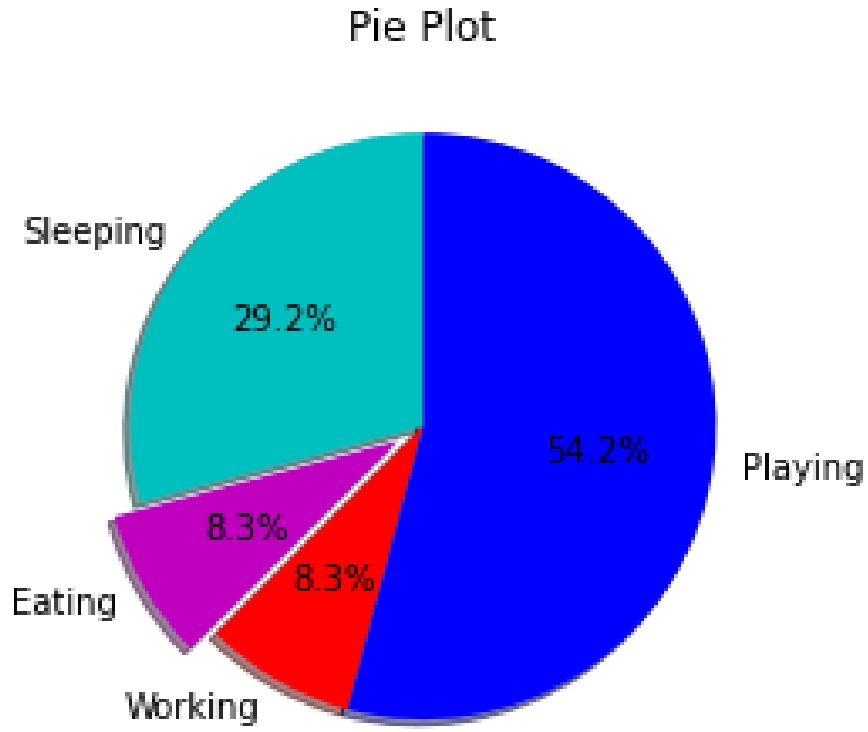
Pie Plot

- A pie chart is a circular statistical graphic, which is divided into slices to illustrate numerical proportion.

```
slices =  [7,2,2,13]
activities = ['Sleeping','Eating','Working','Playing']
cols = ['c','m','r','b']

plt.pie(slices,
        labels=activities,
        colors=cols,
        startangle=90,
        shadow=True,
        explode=(0,0.1,0,0),
        autopct='%1.1f%%')

plt.title('Pie Plot')
plt.show();
```



https://matplotlib.org/tutorials/introductory/sample_plots.html